REVISION 2021 • PRINTING TECHNOLOGY • SEMESTER 4

Print Finishing & Conversion Techniques

A professional diploma handbook for Course Code 4102. It explains book binding, loose-leaf, perfect, case and account book binding, stitching, sewing, folding, forwarding, finishing and modern post-press machines.

4102
Course code

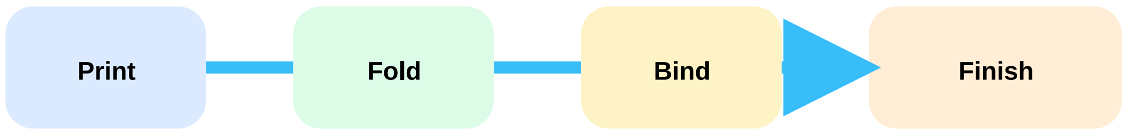
4
Credits

60
Periods

Program Core
Category

Print finishing □□□□□□□ printed sheet customer-ready product □□□□□□□□□□□ post-press stage □□□. Binding style, folding accuracy, trimming, laminating, foil stamping, die cutting □□□□□□□ quality, durability, appearance □□□□□□□ □□□□□□□□□ □□□□□□□□□□□□□□.

Post-press production flow



Course overview

Objectives

Provide basic knowledge on book binding and print finishing, explain need of finishing in press work, familiarize book binding aspects and identify post-press practices and machineries.

Prerequisite

Paper sizes and paper grades from Materials for Printing & Packaging, Semester 3.

Course outcomes

CO	Outcome	Hours
CO1	Explain role and necessities of book binding and print finishing.	15
CO2	Explain key features and operations of book binding.	14
CO3	Explain forwarding and finishing operations of post-press.	15
CO4	Describe features and advancements of post-press machines.	14

Source / syllabus information

Item	Official information
Programme	Diploma in Printing Technology
Code / title	4102 - Print Finishing & Conversion Techniques
Semester / credits	Semester 4 / Credits 4
CO-PO mapping	CO1 to CO4 moderately mapped to PO1.
Model paper	Not specified in supplied syllabus/source.

Redesigned syllabus roadmap

M1 Binding basics

Draw binding classification, explain loose leaf, perfect, case and account binding. Common mistake: mixing binding type with finishing material.

M2 Book operations

Draw folding schemes, sewing styles and end paper positions. Focus: stitching, sewing, folding, binding styles.

M3 Forwarding / finishing

Explain sequence from cutting to casing-in, embossing, foil stamping, laminating, varnishing, die cutting and edge decoration.

M4 Machines and QC

Explain cutter, folder, stitcher, gathering, laminating, perfect binding, case making and quality control checks.

Module 1 - Book Binding and Print Finishing Fundamentals

CO1 • 15 hours • Introduction, binding types, tools, equipment and materials.

Book binding and finishing

Book binding assembles leaves or sections into a usable book. Print finishing improves appearance, protection, handling and commercial value after printing.

Classification

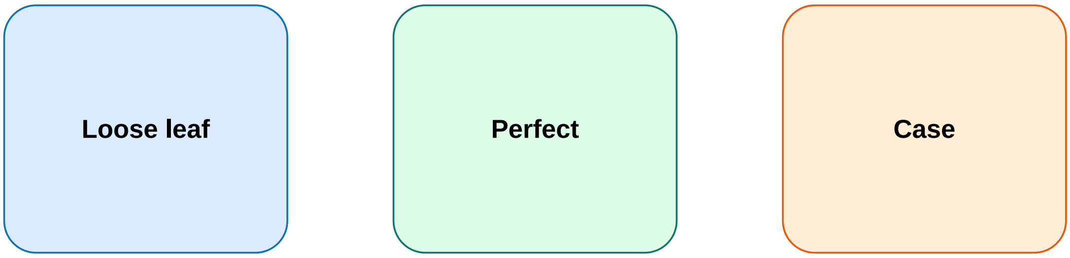
Stationery binding and letterpress binding are major groups. Loose-leaf binding includes spiral, ring, thong, post and comb binding.

Perfect and case binding

Perfect binding uses adhesive on spine, suitable for magazines and paperbacks. Case binding uses hard cover/case, giving strength and premium finish.

Materials

Covering materials, reinforcing materials, securing materials and adhesives decide durability. Finishing materials include eyelets, foils and coloring materials.



Book binding classification examples.

- Define book binding and print finishing.
- Explain loose leaf binding types.
- Compare perfect binding and case binding.
- List binding and finishing materials.

Module 2 - Stitching, Sewing, Folding and End Paper

CO2 • 14 hours • Book binding operations and styles.

Stitching and sewing

Stitching uses wire or thread for quick assembly. Side stitching passes through side margin; saddle stitching passes through centre fold. Sewing gives stronger book blocks: overcast, all-along, French, tape, raised cord, recessed cord and two-on sewing.

Paper sizes and folding

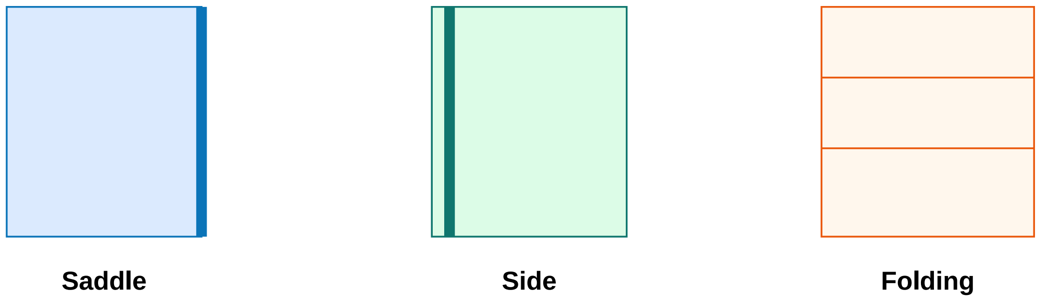
Folio, quarto, octavo, 16mo, 32mo and 64mo represent folding formats. Right angle/French fold changes direction; parallel folding includes zig-zag, letter fold and over-and-over fold.

Binding styles

Quarter bound, three-quarter bound, half bound and full bound books differ by cover material distribution on spine, corners and sides.

End papers

End papers connect book block to cover and improve strength/finish. Types include self, single, double, made, reinforced, cloth joint and zig-zag end paper.



Module 3 - Forwarding and Print Finishing Operations

CO3 • 15 hours • Sequence, purpose and working principles.

Pre-forwarding and forwarding

Major operations are cutting, trimming, folding, gathering, collating, sewing, rounding, backing, head banding, casing-in, creasing, slitting, punching and drilling.

Decorative finishing

Embossing raises image, debossing depresses it, foil stamping transfers metallic foil using heat and pressure, ruling and lettering add functional print features.

Protection and conversion

Laminating, coating and varnishing protect printed surface. Index cutting, inserting, tipping-in, numbering, perforating and die cutting convert the job into required form.

Edge decoration

Edge coloring, painting, sprinkling, gauffering, marbling and edge gilding improve premium appearance.

Module 4 - Post-Press Machines and Quality Control

CO4 • 14 hours • Machine functions, advanced features, modernization and QC.

Major machines

Programmable cutting machine, folding machine, wire stitching machine, rounding and backing machine, gathering machine, laminating machine, coating machine, bundling machine, perfect binding machine and case making machine.

Advanced features

Programmable settings, automated feeding, sensor-based alignment, digital counters, safety interlocks, adjustable guides and quick job changeover improve productivity.

Quality control

Check sequence, page order, trim size, fold accuracy, stitch position, glue strength, cover alignment, lamination bubbles, foil registration and die-cut accuracy.

Troubleshooting

Common faults: wrong sequence, skew folding, loose stitch, adhesive failure, cover misalignment, poor lamination adhesion and burr in cutting.

Rule bank / procedure bank

Perfect binding rule

Spine preparation + correct adhesive + clamping pressure decides pull strength.

Folding rule

Wrong grain direction or wrong fold setting causes cracking and misregister.

QC rule

Never inspect only appearance; check function, durability and dimension.

Practice and quick revision

- One-mark: Define casing-in.
- Short answer: List end paper types.
- Three-mark: Explain saddle stitching and side stitching.
- Descriptive: Explain forwarding operations in correct sequence.
- Application: Select finishing operations for a premium hardbound book.

Objective questions

1. Perfect binding mainly uses

A. Adhesive B. Screen C. Anilox D. Laser

Answer: A

2. Raised image is produced by

A. Debossing B. Embossing C. Slitting D. Gathering

Answer: B

Fresh model question paper

Course Code: 4102
Course: Print Finishing & Conversion Techniques
Time: 3 Hours | Maximum Marks: 100

Part A
1. Define book binding.
2. List four finishing materials.
3. What is saddle stitching?
4. Name any four folding styles.
5. Define end paper.
6. What is embossing?
7. State the purpose of laminating.
8. Name four post-press machines.
9. What is quality control in post-press?
10. Define die cutting.

Part B
11. Classify book binding and explain loose leaf binding.
12. Compare perfect binding and case binding.
13. Explain stitching and sewing processes.
14. Explain paper sizes and folding schemes.
15. Explain end paper types and uses.
16. Explain pre-forwarding and forwarding operations.
17. Explain major print finishing operations.
18. Explain programmable cutting and folding machines.
19. Explain perfect binding machine and case making machine.
20. Explain quality control checks in post-press.

Part C
21. Select suitable binding and finishing method for a hardbound premium book.
22. Prepare troubleshooting notes for folding, stitching and lamination defects.

Complete answer guide

Key answer pattern

Start with definition, give classification, draw neat workflow or machine sketch, explain sequence, list advantages/limitations, applications and QC checks.

Model answer points

For hardbound premium book: use section sewing, rounding/backing, case binding, end papers, laminating/varnish, foil stamping, embossing and edge decoration if required.

References

D. Mendiratta, Binding & Finishing; Spiers Hugh, Introduction to Printing and Finishing; G. Martin, Finishing Process in Printing; Alex J. Vaughan, Modern Book Binding; Ralp Lyman, Binding and Finishing; Alfred Furler, Folding in Practice; P. Kipphan, Handbook of Print Media.